

The 8-Tuple Basis of $\text{omi} \dashv \text{imo} \text{?} 0_o$

Core Claim

The notation:

$\text{omi} \dashv \text{imo} \text{?} 0_o$

is the readable alias of a folded/unfolded 8-tuple relation.

The folded carrier is:

$\text{omi} \dashv \text{imo}$

The unfolded witness surface is:

$\text{?} 0_o$

The computational invariant beneath both is:

$(Q, \Sigma, L, R, \delta, s, t, r)$

Therefore:

$\text{omi} \dashv \text{imo} = (Q, \Sigma, L, R, \delta, s, t, r)$
 $\text{?} 0_o = (Q \ \Sigma \ L \ R \ \delta \ s \ t \ r)$

And the witness relation is:

$\text{omi} \dashv \text{imo} \text{ XOR } \text{?} 0_o$

This means:

folded carrier XOR unfolded readable surface = relation witness

1. The 8-Tuple

The OMI 8-tuple is:

Q Σ L R δ s t r

Meaning:

Q state space / identity context
 Σ alphabet / allowed symbols
L left / source / beginning
R right / target / ending
 δ transition / transformation
s start value
t accept / success / terminal output
r reject / failure / alternative

This is the same shape as a two-way deterministic finite automaton, but OMI uses it as a general relation carrier.

Canonical sentence:

The 8-tuple is the basis of the OMI relation frame.

2. Address Cascade Mapping

The OMI address cascade has eight 16-bit scope/ruler fields:

S0 S1 S2 S3 S4 S5 S6 S7

These map directly to the 8-tuple:

S0 = Q
S1 = Σ
S2 = L
S3 = R
S4 = δ
S5 = s

S6 = t
S7 = r

So the address cascade is:

S0-S1-S2-S3/S4/S5/S6/S7

read as:

Q- Σ -L-R/ δ /s/t/r

or compactly:

(Q, Σ , L, R, δ , s, t, r)

3. Register Compression

The second half of the OMI frame compresses the 8-tuple into four paired 32-bit fields:

CAR
CDR
PAYLOAD
MASK

The pairing is:

CAR = Q & Σ
CDR = L & R
PAYLOAD = δ & s
MASK = t & r

Thus:

Q, Σ → CAR
L, R → CDR
 δ , s → PAYLOAD
t, r → MASK

The full OMI cell is therefore:

```
8 × 16-bit address/ruler fields
+
4 × 32-bit paired relation fields
```

This is not two unrelated halves.

It is:

```
8 individual coordinates
+
4 paired relation carriers
```

4. Null Ring Foundation

The folded null seed is:

```
(NULL . NULL)
```

The byte ring is:

```
(0x00 . 0x20)
(0x20 . 0x7F)
(0x7F . 0xFF)
(0xFF . 0x00)
```

The dot relation is XOR:

```
0x00 ^ 0x20 = 0x20
0x20 ^ 0x7F = 0x5F
0x7F ^ 0xFF = 0x80
0xFF ^ 0x00 = 0xFF
```

The full witness closure is:

```
0x20 ^ 0x5F ^ 0x80 ^ 0xFF = 0x00
```

Canonical law:

(NULL . NULL) is the folded null seed.
The byte ring is the unfolded null cycle.
The dot relation is XOR.
The full cycle closes to zero.

5. Hopf-Style Projection Language

The eight OMI address fields form an 8-coordinate carrier:

S0..S7

After normalization, an 8-coordinate real carrier may be read as sphere-like S^7 data.

A paired OMI/register frame gives sixteen coordinates:

S0..S7
+
CAR/CDR/PAYLOAD/MASK as paired expansion

After normalization, the sixteen-coordinate carrier may be read as a Hopf-style S^{15} total space.

The projection back to an eight-coordinate readable/base surface is the Hopf-style reading:

S^7 fiber
 S^{15} total carrier
 S^8 base/projection horizon

OMI does not need to claim that every address literally is a smooth Hopf fibration.

The precise claim is:

The Hopf vocabulary names the projection pattern:
a paired high-dimensional carrier projects to a lower-dimensional readable
relation surface while preserving the relation witness.

6. Why This Matters

The 8-tuple explains why OMI can compose relations.

Every relation must answer:

```
Q  what state space?  
Σ  what symbols?  
L  what source?  
R  what target?  
δ  what transition?  
s  what start?  
t  what success?  
r  what failure?
```

That means every OMI relation has enough structure to be:

```
framed  
routed  
validated  
witnessed  
projected  
receipted
```

without pretending that notation alone executes the relation.

Canonical boundary:

```
The 8-tuple frames relation.  
The Null Ring closes relation.  
Tetragrammatron validates relation.  
Metatron scribes relation.  
Receipt records relation.
```

7. Final Canon

```
omi---imo is the folded 8-tuple carrier.  
  
?0_o is the readable witness surface.  
  
S0..S7 carry Q, Σ, L, R, δ, s, t, r.
```

CAR, CDR, PAYLOAD, and MASK pair those eight coordinates into four relation fields.

The Null Ring supplies closure.

Hopf language describes the projection from paired carrier space to readable relation surface.

OMI composition is therefore not arbitrary syntax.

It is 8-tuple relation composition under bounded place-value closure.